

MoveX Cab Application - Comprehensive Report

Detailed Explanation, Workflow, and Features

1. Introduction

MoveX is a comprehensive cab booking platform designed to provide seamless mobility solutions. The system consists of three distinct components: the Customer App for riders, the Driver App for driver partners, and the Admin Dashboard for operations management.

This report outlines the complete workflow of the MoveX platform, detailing the technologies used, core features across its components, and the end-to-end booking process.

2. Technologies Used

The MoveX platform is built using a modern, scalable technology stack:

1. Frontend (Mobile Apps): React Native with Expo. React Navigation is used for routing, and Context API for state management. Expo provides native capabilities like location tracking and push notifications.
2. Frontend (Admin Panel): React.js (or integrated within the React Native web capabilities).
3. Backend API: Node.js and Express.js, providing RESTful endpoints for CRUD operations and business logic.
4. Database: MongoDB (with Mongoose ODM) for flexible schema design handling geospatial data and complex relationships.
5. Real-time Communication: Socket.io for live driver tracking, ride matching, and instant status updates.
6. Authentication: Firebase Authentication (supporting Google Sign-In and Email/Password flows).
7. Cloud Hosting: Render for continuous deployment of the Node.js backend API and MongoDB for the database.
8. Payment Gateway: Integrated Razorpay/Stripe (or simulated cash/wallet workflows) for secure transactions.

MoveX Cab Application - Comprehensive Report

Detailed Explanation, Workflow, and Features

3. Core Features

Customer App Features:

- Ride Booking: Intuitive interface to select pickup and drop locations using maps.
- Fare Estimation: Real-time calculation of fares across different vehicle categories (Bike, Auto, Mini, Sedan, SUV).
- Subscription Passes: Tiered pass system allowing users to buy passes (e.g., Bronze, Silver, Gold, Platinum, Diamond) for discounted or free rides up to a certain limit.
- Live Tracking: Real-time visualization of the assigned driver approaching the pickup location on the map.
- Digital Wallet: In-app wallet for seamless top-ups and cashless payments.
- Multilingual Support: English and regional language translations.

Driver App Features:

- Online/Offline Toggle: Drivers can dictate their availability in real-time.
- Ride Acceptance: Incoming ride requests via push notifications and live socket events, with a countdown timer.
- Earnings Dashboard: Real-time tracking of completed rides and accumulated earnings.
- Document Verification: Secure upload of KYC documents (Aadhaar, License, RC, Insurance).

Admin Dashboard Features:

- User & Driver Management: Full CRUD operations to manage profiles, verify driver documents, and approve drivers.
- Dynamic Configuration: Ability to configure base fares, per-km rates, and commission percentages dynamically.
- Pass Management: Create and update subscription pass tiers with customized benefits.

4. Complete Workflow

The end-to-end ride booking workflow is seamlessly managed through a combination of REST APIs

MoveX Cab Application - Comprehensive Report

Detailed Explanation, Workflow, and Features

and WebSockets:

Step 1: Registration and Authentication

Users and Drivers register via Firebase Auth. Drivers submit their documents, which go into a 'pending' state until verified by the Admin. Once approved, the driver can toggle their status to 'Online', which establishes a Socket.io connection.

Step 2: Ride Request

A Customer enters their destination. The backend calculates fare estimates based on distance and applies active pass discounts. Upon confirming the booking, the backend queries MongoDB using Geospatial queries (\$geoWithin) to find online, approved drivers within a specific radius matching the vehicle type.

Step 3: Driver Matching & Dispatch (Sequential Dispatch)

The matching engine sorts available drivers by distance and dispatches the request to them sequentially via Socket.io. If a driver accepts, the ride status transitions to 'accepted'. If they decline or the request times out, it moves to the next closest driver.

Step 4: Active Trip & Live Tracking

The driver receives a 4-digit OTP from the customer to start the ride. During the trip, the driver's app continually emits location updates via Socket.io, allowing the customer to track the vehicle live on their map.

Step 5: Completion and Payment

Upon reaching the destination, the driver marks the ride as complete. Fare calculations finalize (accounting for wallet balances or cash). Both parties are then prompted to submit a rating and review, concluding the workflow.

5. Conclusion

MoveX Cab Application - Comprehensive Report

Detailed Explanation, Workflow, and Features

The MoveX Cab application represents a robust, highly scalable ride-hailing solution. By leveraging Node.js for high-concurrency APIs, MongoDB for rapid geospatial queries, and Socket.io for low-latency live tracking, the architecture guarantees a smooth and responsive experience for both customers and drivers.